

Cosmic Mesh: Decentralized Node Architecture

The Cosmic Mesh is an experimental peer-to-peer network architecture designed to maintain high-speed connectivity in remote environments without relying on centralized ISPs.

By utilizing a specialized gossip protocol for node discovery and employing AES-256 for all end-to-end encryption, the network ensures that packet sniffing and man-in-the-middle attacks are mathematically impossible.

System Specifications & Capabilities:

- Routing: Utilizes self-healing routing paths using the B.A.T.M.A.N. algorithm.
- Latency: Highly optimized to maintain under 50ms ping across a maximum of 10 node hops.
- Frequencies: Operates primarily on 900MHz LoRa bands for extended long-range transmission.
- Hardware: Nodes run on low-power ARM Cortex-M4 processors consuming less than 2 watts.

This documentation is strictly for Hackweek technical evaluation purposes. Do not deploy the Mesh protocol in unauthorized spectrums.